
# Subash Pandey

**Date of birth:** 14/08/2000 | **Place of birth:** Kathmandu, Nepal | **Nationality:** Nepalese |

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## ABOUT ME

I'm an AI/ML engineer from Nepal with a strong interest in Generative AI and applied machine learning. I enjoy exploring the intersection of research and implementation, and I'm driven by curiosity, clarity, and continuous learning. I take pride in writing clean code, thinking deeply about problems, and building meaningful, responsible AI solutions that create real impact.

## WORK EXPERIENCE

**REMOTE AI/ML ENGINEER – SCOPIC SOFTWARE LLC** – 23/10/2024 – Current – KATHMANDU, NEPAL

**Website:** https://scopicsoftware.com/

- Contributed to the development and maintenance of an AI Receptionist platform; responsible for implementing and maintaining AI-driven features, resolving production bugs, and ensuring system reliability and conversational quality across deployments.
- Led the development and deployment of a GenAI sales chatbot integrated with Zoho CRM, using LangGraph, RAG pipelines, and Qdrant to enable lead qualification, intent detection, user info extraction, lead scoring, and SEO tracking (referrer, session time, last visited page).
- Conducted R&D on extracting audio features (pitch, pause, pace, etc) using tools such as Parselmouth, Librosa, and Praat, supporting internal AI capabilities for voice-based applications.
- Estimated AI project scope and delivery timelines by breaking down technical tasks into phases with man-hour granularity; supported both internal initiatives and client-facing projects.
- Maintained and expanded the AI service layer of an interviewer platform, responsible for various types of question generation, grading logic, prompt engineering, and end-to-end GenAI integration.
- Implemented evaluation pipelines to assess chatbot and RAG performance, improving grounding and response reliability. Performed rigorous evaluations of GenAI output quality (e.g., RAG hallucination rate, chatbot response accuracy, cost, latency), continuously improving relevance and grounding fidelity.
- Collaborated with fully remote, cross-functional teams while following strong Git practices, agile workflows, and internal security policies; worked closely with non-technical stakeholders to translate business goals into deliverable AI features.
- Contributed to the development of a keyword research and SEO SaaS platform as a full stack developer; built and maintained backend services, managed database schemas and migrations, integrated subscription billing, implemented authentication flows, and developed AI-driven features for keyword analysis and content insights.
- Tech stack includes: Python, LangGraph, LangFuse, TypeScript, Node.js, FastAPI, Docker, AWS (Lambda, Fargate, OpenSearch), GCP (Cloud Run), Serverless, databases(redis, mysql), jinja2 and others.

**AI/ML ENGINEER – PEACE NEPAL DOT COM** – 01/07/2024 – 30/11/2024 – KATHMANDU, NEPAL

**Website:** https://peacenepal.com/

Main Activities
- Design and develop chatbots for specific use cases like Banking, Travel and Customer Support.
- Implement Retrieval-Augmented Generation (RAG) and Multi-agent framework for enhanced responses.
- Integrate databases and knowledge sources for comprehensive answers.
- Deploy interactive web chat applications.

Responsibilities
- Plan and manage chatbot development projects.
- Fine-tune LLMs for domain-specific knowledge and needs.
- Conduct testing and optimisation for accuracy and performance.
- Maintain documentation and report on project progress.

**AI/ML ENGINEER – ICEBRKR AI SOLUTIONS.** – 22/03/2024 – 14/06/2024 – KATHMANDU, NEPAL

- Engineer real-world AI solutions using machine learning.
- Build scalable ML models for data analysis and automation.

- Optimise and deploy models for production in cloud environments.
- Collaborate with cross-functional teams (data science, engineering) to deliver impact.
- Proactively learn and apply cutting-edge AI/ML or frameworks and tools ( HuggingFace, PyTorch, AWS).

**DATA ANALYST – CONTENTIO LAB PVT. LTD.** – 31/08/2021 – 28/04/2022 – KATHMANDU, NEPAL

- Analysed large datasets using Python to identify trends and patterns, fast-tracked the process of data wrangling.
- Created clear and concise data visualisations (charts, dashboards) using Python libraries (e.g., Matplotlib, Seaborn) that effectively communicated insights to stakeholders.
- Developed and maintained robust data pipelines using Python scripting to ensure data accuracy and efficiency, leading to a 20% reduction in data processing time.
- Supported data-driven decision making across various departments (marketing, sales, etc.) by providing Python-powered analysis (e.g., customer segmentation, churn prediction), leading to cost-saving measures through market analysis.
- Proficient in data analysis tools (e.g., SQL, Python libraries like Pandas, NumPy) and comfortable working with large and complex datasets.

**FRONTEND INTERN – IMARK PRIVATE LIMITED** – 10/10/2020 – 25/01/2021 – KATHMANDU, NEPAL

- Developed a fully functional to-do web application using React.js, featuring functionalities like task creation, management, and completion.
- Implemented a CRUD (Create, Read, Update, Delete) backend to support data persistence for the to-do application, demonstrating understanding of data interaction.
- Integrated a secure login component using React for user authentication.

**TUTOR – BUDHANILKANTHA EDUCATION SERVICES** – 07/10/2018 – 04/04/2019 – KATHMANDU, NEPAL

- Tutored A-Level Computer Science, specialising in the syllabus curriculum.
- Helped failing students to obtain passing grades.
- Developed personalised lesson plans aligned with A-Level topics using personal notes.
- Provided targeted exam preparation strategies based on A-Level assessment format.

## EDUCATION AND TRAINING

26/09/2022 – 17/12/2023 Exeter, United Kingdom

### MASTER OF SCIENCE IN DATA SCIENCE University of Exeter

Main Subjects:

- Statistics and Data Modelling: Statistical Data Modelling, Fundamentals of Data Science, Stochastic Processes
- Machine Learning: Machine Learning, Learning from Data
- Social Network Analysis: Social Networks and Text Analysis
- Data Science Applications: Introduction to Data Science, Digital Business Models

Occupational Skills:

- Data Analysis: Ability to clean, manipulate, and analyze large datasets.
- Machine Learning: Ability to design, build, and deploy machine learning models.
- Data Visualization: Ability to create clear and informative visualizations of data.
- Communication: Ability to communicate complex data insights to both technical and non-technical audiences.
- Problem-solving: Ability to identify and solve problems using data-driven approaches.
- Business Acumen: Understanding of how data science can be applied to solve business problems and create new opportunities.

**Website** https://www.exeter.ac.uk/ |

**Field of study** Inter-disciplinary programmes and qualifications involving natural sciences, mathematics and statistics |

**Final grade** Merit | **Level in EQF** EQF level 7 | **National classification** 7 | **Type of credits** ECTS | **Number of credits** 90 |

**Thesis** Machine Learning the Steam Video Game Database

12/08/2019 – 10/12/2021 London, United Kingdom

**BACHELOR OF SCIENCE (HONORS) COMPUTING** London Metropolitan University

Main Subjects:
- Computer Science Fundamentals: Computer Hardware and Software Architectures, Fundamentals of Computing, Logic and Problem Solving, Programming
- Software Development: Advanced Programming and Technologies, Software Engineering, Application Development, Project
- Database Management: Databases
- Networking: Network Operating Systems
- Emerging Technologies: Cloud Computing and the Internet of Things, Mobile Application

Occupational Skills Gained:
- Programming: Proficiency in programming languages (Python and Java).
- Software Development: Ability to design, build, test, and deploy software applications.
- Problem-solving: Applying logic and programming skills to solve complex problems.
- Database Management: Understanding and managing databases for data storage and retrieval (MySQL).
- Networking Concepts: Knowledge of network infrastructure and operating systems.
- Project Management: Experience in planning, executing, and delivering a software development project.
- Professionalism and Ethics: Understanding of professional conduct and ethical considerations in the computing field.

**Website** https://www.londonmet.ac.uk/ | **Field of study** Field unknown | **Final grade** 2:1 |

**Thesis** Floki - An eCommerce Web Application

## PROJECTS

### Activity Recognition using Accelerometer and Gyroscope Data

This project explored the use of XGBoost for classifying human physical activities using sensor data from smartphones and watches (data courtesy of external sources). The project aimed to achieve high accuracy in activity recognition using machine learning.
- Data Acquisition: Leveraged existing datasets containing accelerometer and gyroscope data collected from phones and watches during various activities.
- Data Preprocessing: Techniques like cleaning, normalization, and feature engineering were applied to prepare the sensor data for XGBoost modeling.
- XGBoost Model Development:
  - Hyperparameter Tuning: Employed random search to optimize the hyperparameters of the XGBoost model, ensuring optimal performance. Optimized XGBoost model with random search for high accuracy (>85%).
  - Model Training: Trained the XGBoost model on the preprocessed data for accurate activity classification.
  - Model Evaluation: Performance metrics like accuracy, precision, and recall were used to assess the effectiveness of the model.
- Activity Recognition: Classified physical activities (walking, jogging, stairs, etc.) using XGBoost on sensor data (accelerometer, gyroscope) from phones/watches (existing datasets).
- Activities Classified: Walking, Jogging, Stairs, Sitting, Standing, Typing, Brushing Teeth, Eating (Soup, Chips, Pasta), Drinking, Eating (Sandwich), Writing, Clapping, Folding Clothes.

**Link** https://github.com/notsubash/Activity-Recognition

01/03/2023 – 25/07/2023

### Machine Learning the Steam Video Game Database

This project utilised network analysis to explore the evolution of game genres on the Steam platform. Key findings included:
- Emergent Sub-genres: Identified sub-genres evolving from popular categories like "Action," "Indie," and "Racing," highlighting genre diversification within the gaming market.
- Influential Genres: Pinpointed "Indie" and "Casual" genres as significant influencers within the Steam network, suggesting their impact on broader genre trends.
- Player Engagement Drivers: Revealed game ratings, release year, in-game achievements, and lower prices/free-to-play models as key factors driving player engagement.
- Societal and Economic Impact: Examined how genres like "Massively Multiplayer" and "Free to Play" contribute to cultural exchange, communication, and the overall growth of the gaming industry.

**Links** https://github.com/notsubash/Game-Genre-Network-Analysis | https://www.youtube.com/watch?v=IjWUKT-kXw4&t=23s

### Wikipedia Adminship Network Analysis

This project investigated the social network dynamics within Wikipedia's administrator election process using network analysis techniques. NetworkX, a Python library, was employed to analyze the relationships between users participating in these elections.

- Data Acquisition: Extracted data on user voting behavior and relationships from historical Wikipedia administrator election records.
- Network Construction: Constructed a network where users were represented as nodes and their voting interactions formed the edges.
- Centrality Measures: Calculated centrality measures (e.g., degree centrality, closeness centrality, betweenness centrality) to identify influential users within the administrator election network.
- Clustering: Performed community detection algorithms to identify groups using hierarchical clustering (clusters) of users with similar voting patterns.
- Analysis: Analysed the results to understand the structure of the administrator election network, identify key players, and explore potential voting blocs.

**Links** https://www.youtube.com/watch?v=DywRZz8iRAo | https://github.com/notsubash/WikiVoteNetworkAnalysis

### Stock Variables Analysis

This project utilized exploratory data analysis techniques to gain insights into factors influencing stock prices across different exchanges.

- Data Exploration: Analyzed historical stock data to identify trends and patterns in closing prices across various exchanges.
- Feature Engineering and Correlation Analysis: Investigated relationships between various stock attributes (e.g., volume, P/E ratio, dividend yield) and their potential impact on closing prices. This may have involved techniques like feature engineering and correlation analysis.
- Variable Importance: Identified the most significant variables that contribute to stock price fluctuations at the end of the day. This could involve techniques like regression analysis or feature selection algorithms.
- Data Clustering: Clustered the stock data based on shared characteristics of various attributes, potentially revealing hidden patterns or market segments.

**Link** https://github.com/notsubash/stock-variables

### Floki - eCommerce

Developed Floki, a full-stack e-commerce application leveraging the MERN stack (MongoDB, Express.js, React.js, Node.js) and Redux for state management.

- Intuitive Frontend (React): Built a user-friendly and responsive interface for product browsing, searching, filtering, and cart management, utilizing React Router for seamless navigation.
- Robust Backend (Node.js & Express.js): Established secure API endpoints for user authentication, product management, order processing, and payment integration using Mongoose for efficient data interaction with the MongoDB database.
- Scalable Data Management (MongoDB): Designed a scalable database schema with Mongoose to efficiently store product information, user data, and order details.
- Centralized State Management (Redux): Implemented Redux for centralized state management, ensuring data consistency and smooth user interactions across the application.

**Link** https://github.com/notsubash/Floki-eCommerce

## SKILLS

#### Data Science

Graph Networks | Data Science and Analytics | Machine Learning | Gephi | MLOps | Pandas | numpy

#### Presentation and Prototyping

Microsoft Office | Gradio | Tableau | Streamlit

#### Software Development

Python | Project Management | FastAPI | Stripe | TypeScript | Git | Audio Feature Extraction | Node.js

#### Cloud Computing

AWS | GCP | Docker | Serverless

**LLM / AI**

Langchain | LLM | Vector Database | LangGraph | NLP | LLM Finetuning | Prompt Engineering | GenAI Evaluation | LLMOps

**Databases**

Databases | SQL | Redis | MySQL | Postgres | NoSQL

## LANGUAGE SKILLS

Mother tongue(s): **NEPALI**

Other language(s):

| | UNDERSTANDING | | SPEAKING | | WRITING |
|---|---|---|---|---|---|
| | Listening | Reading | Spoken production | Spoken interaction | |
| **ENGLISH** | C1 | C1 | C1 | C1 | C1 |
| **HINDI** | C2 | B2 | B2 | B2 | B2 |

*Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user*

*I am eager to leverage my skills and experience to contribute to a team-oriented environment. I am a highly motivated individual with a strong desire to learn and grow. I am confident that I can make a significant impact on your organisation.*